KUBERNETES + ARGOCD DEPLOYMENT PROJECT

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Course: DevOps

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Project Overview

This project demonstrates container orchestration and continuous deployment using Kubernetes and ArgoCD.

The setup includes:

- Kubernetes cluster creation using Kind
- Application deployment on Kubernetes
- ArgoCD for continuous deployment
- Kubernetes Dashboard for monitoring

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Architecture / Workflow

1. Kubernetes cluster is created using Kind
2. kubectl is used to manage the cluster
3. ArgoCD is installed in the cluster
4. Application is deployed using ArgoCD
5. Services are exposed using port forwarding
6. Kubernetes Dashboard is used for monitoring

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Tools Used

- Kubernetes
- Kind (Kubernetes in Docker)
- kubectl
- ArgoCD
- Docker

Execution Screenshots

- Kind cluster creation
- kubectl output
- ArgoCD UI
- Application running
- Kubernetes dashboard

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Challenges Faced

- Cluster setup issues
- ArgoCD configuration
- Port forwarding issues
- Dashboard access setup

Conclusion

This project demonstrates Kubernetes orchestration and continuous deployment using ArgoCD. It helps automate deployment and manage containerized applications efficiently.

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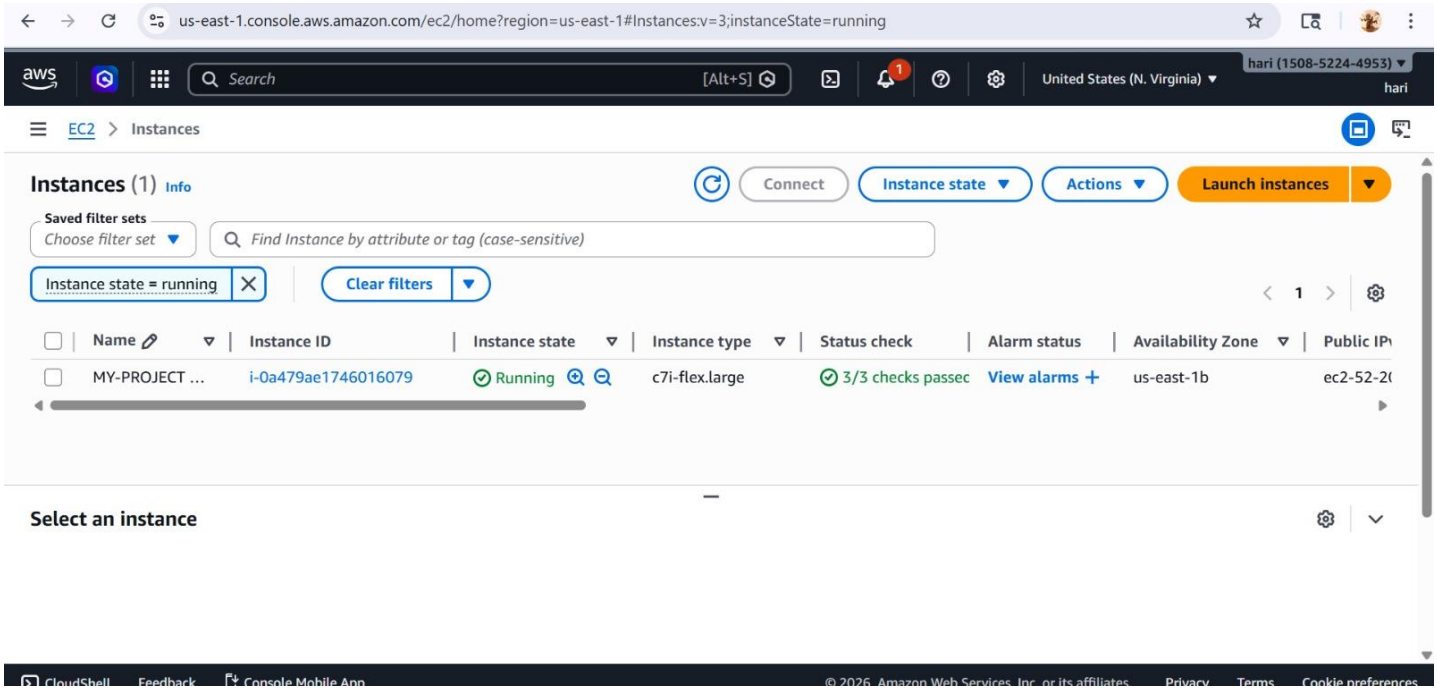

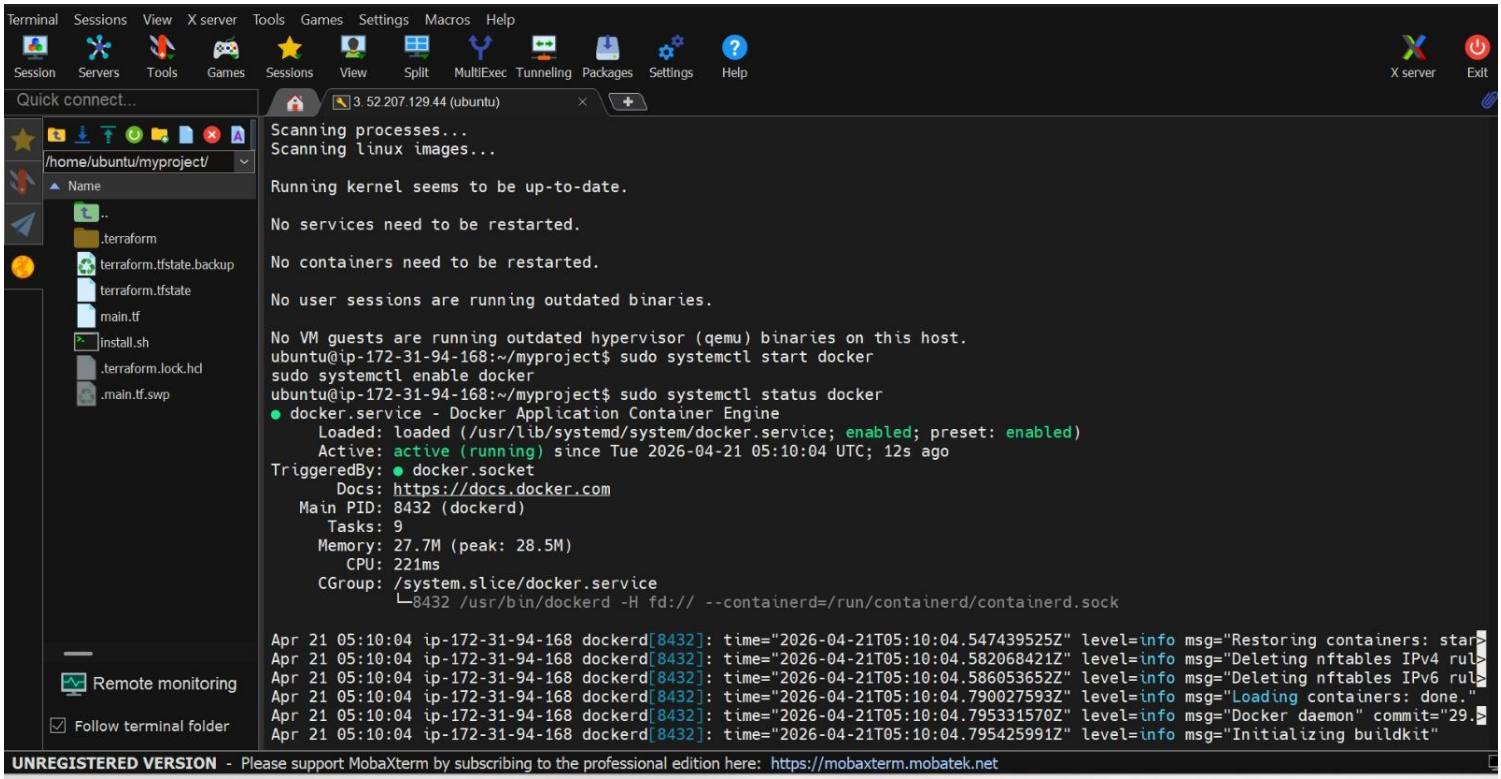

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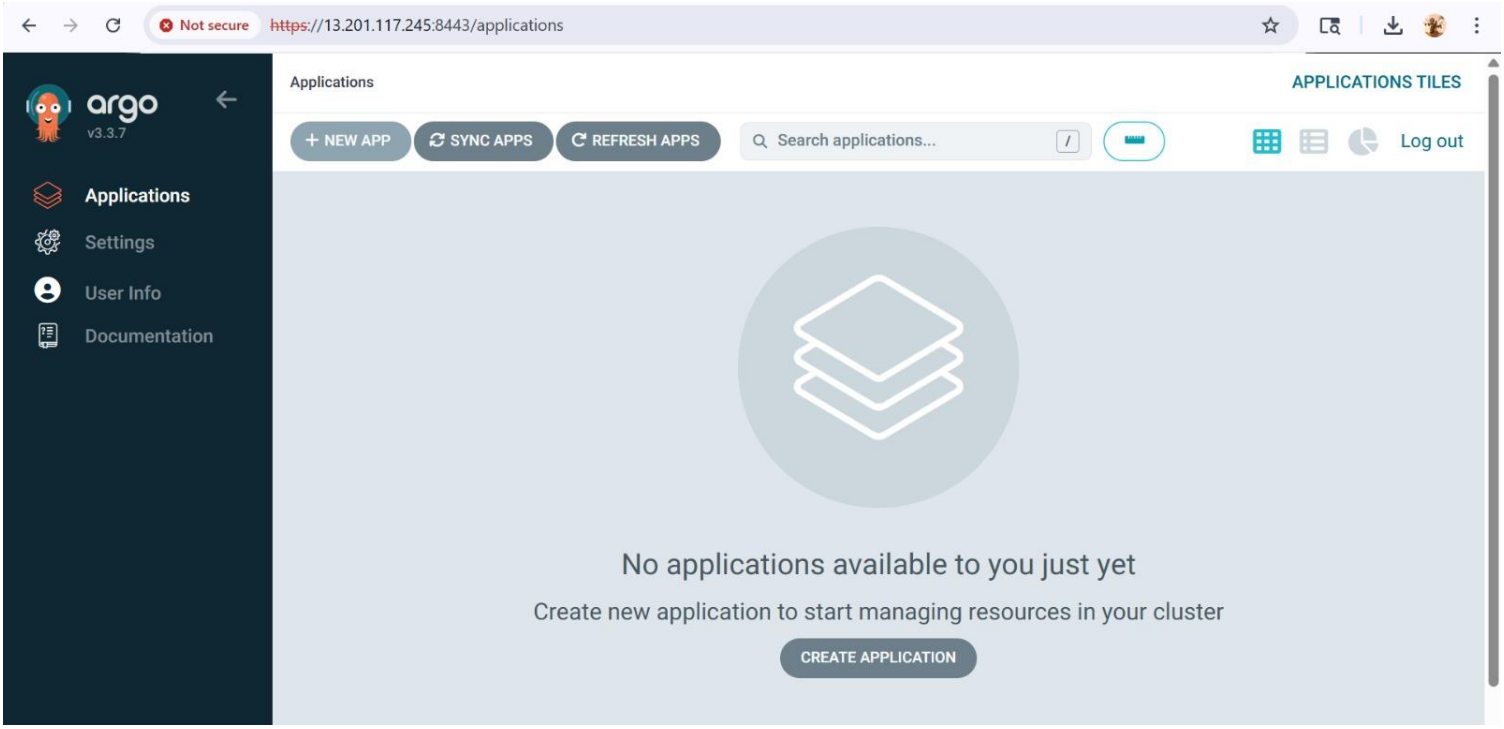

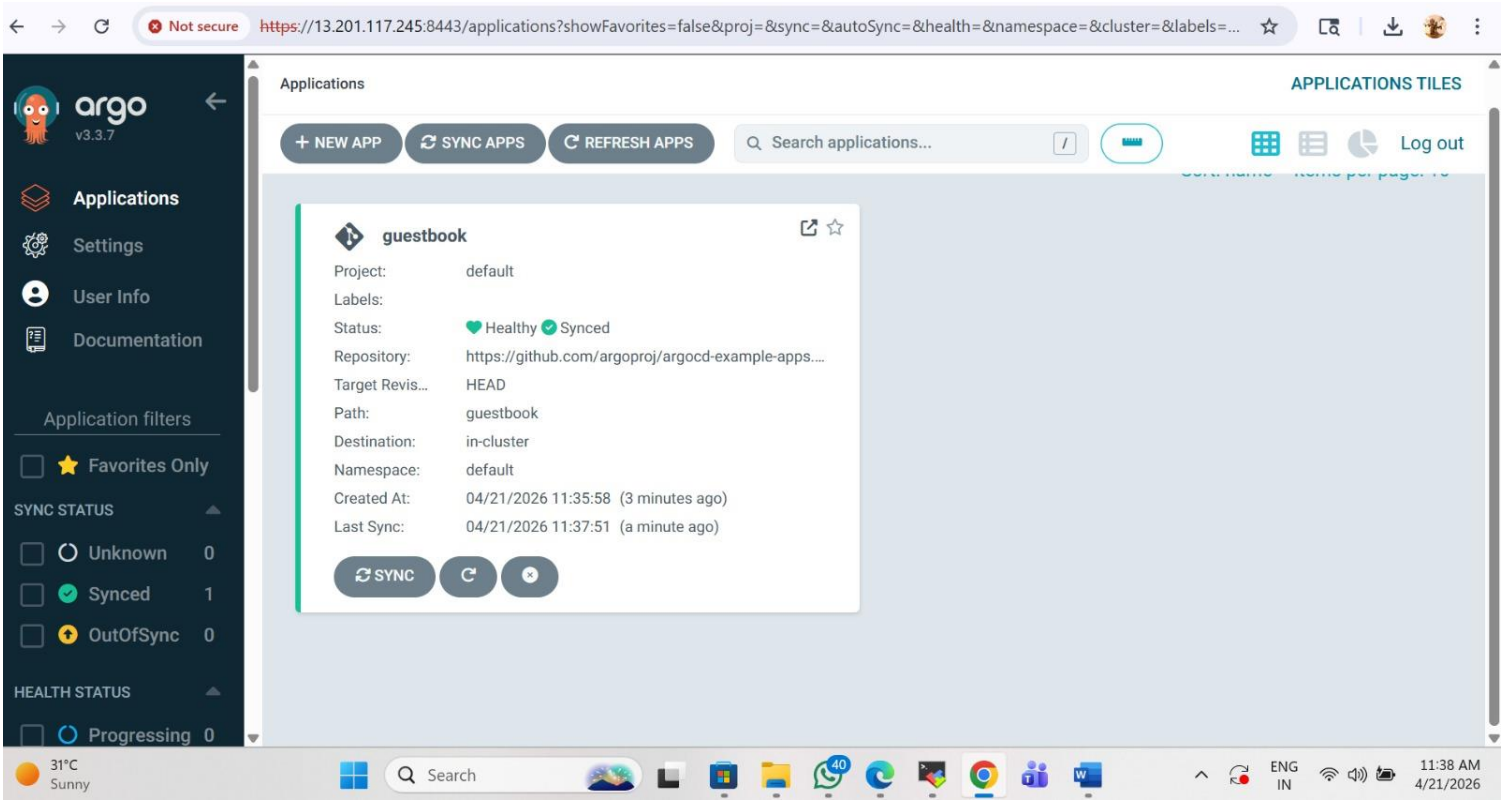

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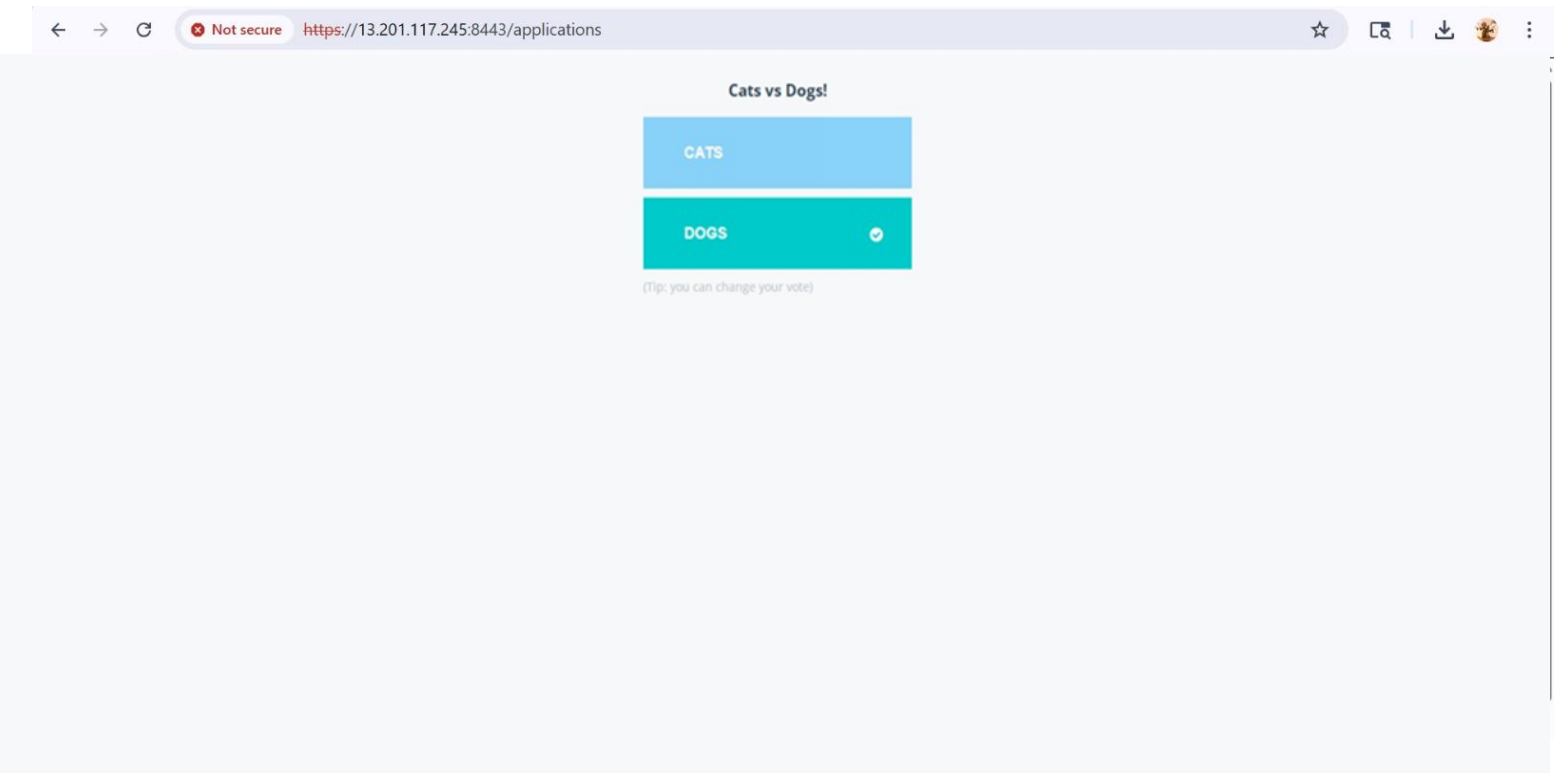
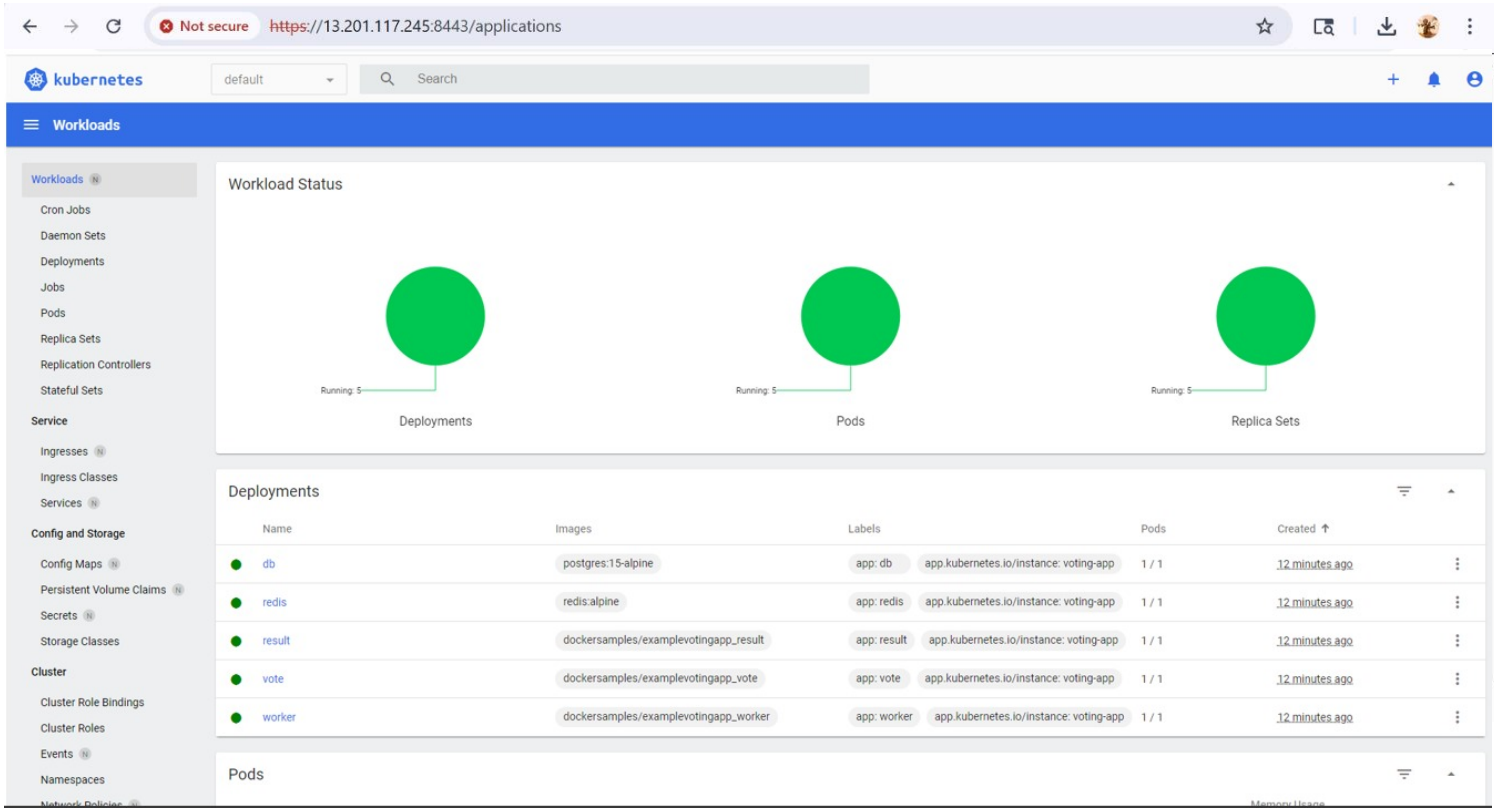
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